

MINI-PROJECT IMPLEMENTATION PLAN

NLP-Based Early Risk Screening for Autism Spectrum Disorder in Toddlers Using Caregiver Observation Reports

Domain: Natural Language Processing / Healthcare AI

Project Goal

Develop a longitudinal caregiver-observation and clinical decision-support prototype that records timestamped free-text and guided observations, extracts ASD-relevant behavioural cues using NLP, builds a cumulative behavioural profile, estimates screening concern using machine learning, and generates separate explainable reports for caregivers and clinicians. The system supports screening and observation; it does not diagnose ASD.

1. Short Project Description

Parents or caregivers record natural-language observations about a toddler over a doctor-advised monitoring period (for example, one week). The application also provides guided observation prompts so that important behavioural domains are observed more systematically and the record is not limited only to behaviours that concern the caregiver. Each entry is timestamped and stored. An NLP service converts the text into structured behavioural events, including the behavioural domain, status/polarity, context, evidence phrase and confidence. Across the monitoring period, these events are aggregated into a longitudinal behavioural profile. A machine-learning decision-support layer estimates screening concern and an explainability layer identifies the evidence supporting the output. An LLM is used downstream to convert structured findings into two different reports: a simple caregiver report and a more detailed clinician report.

2. Scope and Safety Boundary

- The application is an early risk-screening and decision-support prototype, not a diagnostic system.
- The clinician remains the decision-maker; model output must not be presented as a confirmed diagnosis or a clinical probability of ASD.
- Original caregiver text must be retained alongside the NLP interpretation so that clinicians can review the source evidence.
- The system must distinguish 'behaviour not observed' from 'behaviour absent/reduced'.
- For the student prototype, use de-identified or synthetic data. Do not collect identifiable real patient data without the required institutional/ethical approvals and data-governance procedures.

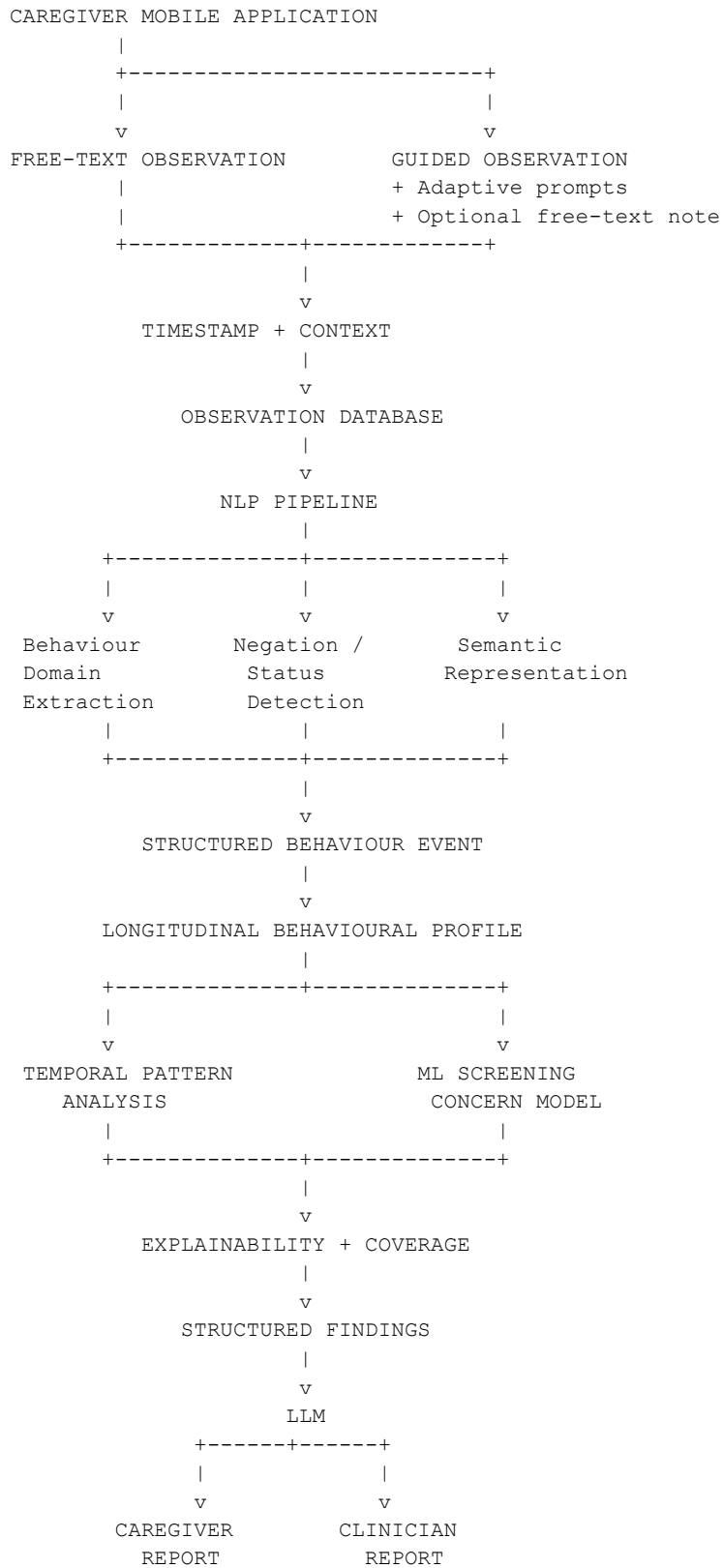
3. Complete List of Proposed Features

Feature	Purpose / Expected Behaviour
User and child profile management	Parent/caregiver authentication, child profile, editable basic information and protected access.
Doctor-defined observation period	Support a configurable monitoring window such as 7 days rather than assuming a fixed duration.

Spontaneous free-text observation diary	Parents can enter observations at any time in their own words.
Guided observation	The app periodically suggests ASD-relevant behaviours to observe, with choices such as observed normally, observed with concern, not observed and not sure, plus an optional free-text note.
Adaptive guided observation	The system identifies behavioural domains with insufficient coverage and recommends future observation prompts for those domains.
Automatic date and timestamp	Every observation is automatically associated with date/time information.
Optional observation context	Playing, eating, social interaction, outdoors, with family, with unfamiliar people, other; context may also be inferred from text.
Longitudinal observation database	Maintain the complete sequence of caregiver observations rather than treating each entry as an isolated sample.
NLP text preprocessing	Normalization, tokenization and appropriate linguistic preprocessing while preserving clinically important negation words.
Behavioural cue extraction	Map free text to domains such as response to name, eye contact, joint attention, gestures, communication, social interaction, pretend play, repetitive behaviour, restricted interests and sensory behaviour.
Negation/status detection	Determine whether a behaviour is present/typical, reduced/concerning, uncertain or not observed.
Evidence phrase extraction	Retain the exact phrase that supports each extracted behavioural event.
Semantic representation	Use TF-IDF baselines and sentence embeddings; DistilBERT may be explored as an advanced extension.
Structured behavioural events	Store domain, status, timestamp, context, evidence phrase and NLP confidence for each analysed observation.
Longitudinal behavioural profile	Aggregate repeated observations by domain, day, status and context.
Temporal pattern analysis	Identify recurrent, inconsistent, context-dependent and isolated behavioural patterns across the observation period.
Observation coverage indicator	Show which behavioural domains have good, limited or insufficient observation coverage.
ML-based screening concern estimation	Use a validated experimental classifier to estimate model-based screening concern; thresholds must be documented and not equated with diagnosis.
Explainable AI	Show influential behavioural domains and evidence phrases; SHAP/feature coefficients may be used where appropriate.
Parent/caregiver report	Simple, non-alarming summary of observations, strengths/typical behaviours, areas worth discussing and recommended next step.
Clinician report	Detailed domain-wise counts, temporal patterns,

	representative evidence, original observations, model output, limitations and observation coverage.
Observation timeline	Chronological view of entries for clinician review.
Behavioural dashboard	Visual summaries of domain coverage, recurring patterns and observation counts.
LLM-based report generation	LLM receives structured findings and produces role-specific reports; it is not used as the primary diagnostic engine.
Report export/share	Generate a shareable summary for review by the healthcare professional, subject to privacy safeguards.
Robustness handling	Test informal language, spelling errors, paraphrases, short entries and negation.
Auditability	Preserve source observation, model interpretation and generated report so outputs can be traced back to evidence.

4. System Architecture



Recommended implementation principle: the NLP/ML layer determines the structured findings and screening-support output; the LLM is used downstream to explain and format those findings. This separation improves traceability and reduces the risk of allowing an LLM to act as an uncontrolled diagnostic engine.

5. Technology Stack

Layer	Recommended Technology	Role
Mobile application	Flutter	Flutter healthcare-app experience.
Authentication	Supabase Auth	Parent/caregiver authentication and secure sessions.
Database	PostgreSQL / Supabase	Child profiles, observations, structured behavioural events, longitudinal aggregates and reports.
Access control	Supabase Row-Level Security	Restrict access to authorised user records.
AI backend	Python + FastAPI	Expose NLP/ML services to the Flutter application through REST/JSON.
Data processing	Pandas, NumPy	Dataset preparation, EDA and experimentation.
Machine learning	scikit-learn	Logistic Regression, SVM, calibration/evaluation utilities.
NLP	spaCy / NLTK	Preprocessing, linguistic rules and optional entity/context extraction.
Text baseline	TF-IDF	Interpretable NLP baseline.
Semantic NLP	Sentence Transformers	Context-aware sentence embeddings.
Advanced transformer (stretch)	Hugging Face DistilBERT	Optional fine-tuning if time and compute permit.
Explainability	SHAP + evidence phrase extraction	Explain model output and link findings to caregiver text.
LLM	API-based or approved local LLM	Structured caregiver and clinician report generation.
Charts/dashboard	Flutter charting package	Observation coverage and longitudinal visualisations.
Version control	GitHub	Collaborative development, branches, issues and integration.

6. Recommended 10-Week Main Implementation Plan

Week	Main Implementation
Week 1	Freeze requirements; study ScreeningPaL and key literature; obtain datasets; perform initial EDA; define behavioural taxonomy; finalise architecture, API contracts and database schema; create Flutter and FastAPI project skeletons.
Week 2	Build structured-data ML baseline; begin synthetic caregiver-narrative generation and validation; implement authentication, child profile, observation-period setup and free-text observation storage. Target first end-to-end API connection.
Week 3	Implement NLP preprocessing, negation handling and initial behavioural cue extraction; implement guided observation UI, timestamp/context capture and observation history; finalise synthetic-text dataset.
Week 4	Train TF-IDF + Logistic Regression/SVM models; implement sentence embeddings; return structured behavioural events through FastAPI; integrate NLP results with PostgreSQL/Supabase.
Week 5	Implement longitudinal behavioural profile, domain-wise aggregation, temporal recurrence/context analysis and observation coverage indicator; begin adaptive guided-observation logic.
Week 6	Integrate selected ML screening-concern model with longitudinal features; implement explainability/evidence phrase highlighting; compare structured baseline, TF-IDF and semantic-embedding approaches.
Week 7	Implement parent report and clinician report schemas; integrate LLM for controlled structured report generation; build timeline and behavioural dashboard views.
Week 8	Complete Flutter-FastAPI-Supabase integration; implement adaptive guided prompts, report export/share prototype and end-to-end workflow; optionally begin DistilBERT experiment.
Week 9	Perform model evaluation, robustness testing, negation/paraphrase/typo tests, error analysis, probability calibration if used, subgroup/fairness checks where sample sizes permit, and security/privacy review.
Week 10	Final system testing and bug fixing; freeze models; prepare final evaluation tables, architecture diagrams, user manual, project report, PPT/poster, demo cases and viva preparation.

7. First Major Milestone - Prototype 1

Target: The objective is to prove the complete technical path from caregiver entry to AI interpretation and persistent storage before building advanced models or reports.

- Parent/caregiver can log in.
- Parent can create/select a child profile.
- A monitoring/observation period can be created.
- Parent can type a free-text behavioural observation.

- The application automatically records date and time.
- Observation is stored in PostgreSQL/Supabase.
- Flutter sends the observation to the Python/FastAPI NLP endpoint.
- NLP identifies at least one behavioural domain and its status/polarity.
- The API returns a structured behavioural event.
- The structured event is stored and linked to the original observation.
- The application displays observation history and the extracted interpretation.

Prototype 1 - Example

Caregiver entry: "He doesn't look towards me when I call his name."

Expected backend response (illustrative):

```
{
  "domain": "response_to_name",
  "status": "concern",
  "negation_detected": true,
  "evidence": "doesn't look towards me when I call his name",
  "confidence": 0.91
}
```

Prototype 1 is considered successful only when this entire chain works end-to-end: Flutter -> Supabase/PostgreSQL -> FastAPI -> NLP analysis -> structured event -> persistent storage -> Flutter display.

8. Suggested Organisation

Primary Work
Dataset/EDA, synthetic narratives, behavioural taxonomy, NLP preprocessing, negation, TF-IDF, sentence embeddings, ML evaluation, longitudinal analytics and explainability.
Flutter UI, authentication, child profiles, observation diary, guided observation, PostgreSQL/Supabase schema, API integration, dashboards and reports.
Integration, testing, GitHub reviews, documentation, demonstration and understanding of the complete system.

9. Recommended Behavioural Domains for the Prototype

- Response to name
- Eye contact
- Joint attention / pointing to share interest
- Gestures
- Social interaction / shared enjoyment
- Communication / language
- Play and pretend play
- Repetitive behaviours
- Restricted interests
- Sensory responses

The taxonomy should be finalised after literature review and clinical consultation. The student team should not invent clinical interpretations independently.

10. Data and Model Development Strategy

- Use a public structured ASD screening dataset as the initial supervised-learning source.
- Establish a structured-data baseline before the NLP model.
- Generate caregiver-style narratives from structured records for NLP experimentation, while validating that generated text does not introduce symptoms absent from the source record.
- Retain separate train, validation and test partitions; prevent variants generated from the same original record from leaking across splits.
- Start with TF-IDF + Logistic Regression/SVM, then sentence embeddings. Treat DistilBERT as a stretch goal.
- Evaluate accuracy, precision, recall/sensitivity, F1-score, ROC-AUC and confusion matrix; do not rely on accuracy alone.
- Perform error analysis on false positives and false negatives and test robustness to spelling errors, informal language, paraphrases and negation.

11. Minimum Final Demonstration Scenario

1. Create/select a child profile and start a 7-day observation period.
2. Enter at least one spontaneous free-text observation.
3. Complete at least one guided observation prompt.
4. Show automatic timestamping and the stored observation timeline.
5. Show NLP extraction of behavioural domain, status and evidence phrase.
6. Show the longitudinal behavioural profile and observation coverage.
7. Show a model-based screening-support result with explanation.
8. Generate and display the caregiver report.
9. Generate and display the clinician report.
10. Demonstrate that the report links back to original caregiver observations and clearly states that the system is not a diagnostic tool.

12. Key Deliverables by the End of the Project

- Working Flutter mobile application
- PostgreSQL/Supabase database and documented schema
- Python/FastAPI NLP/ML service
- Prepared experimental dataset and preprocessing pipeline
- Trained and evaluated baseline/semantic NLP models
- Longitudinal behavioural profiling and guided-observation module
- Explainability/evidence module
- Caregiver and clinician report generator
- Model evaluation and robustness/error-analysis results
- GitHub repository with README and setup instructions
- Final project report, presentation and demonstration cases

13. Success Criteria

The project should be judged not only by classification accuracy but by whether the complete system reliably converts repeated caregiver observations into traceable, clinically relevant behavioural information. A

successful prototype should demonstrate end-to-end operation, preserve source evidence, handle longitudinal observations, avoid unsupported diagnostic claims, and generate useful role-specific summaries.